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FOR MBA, BBA AND HSC
Place : Near Dhamrai Govt. College, Dhamrai, Dhaka

Online Full Course Batch
Special Lecture Sheet
Exam Batch -2025
Subject: Accounting Theory
Chapter -12: Capital Structure

For

- **BBA 4th Year, Accounting Department**

New Edition July 2026

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Chapter 12

Capital Structure [মূলধন কাঠামো]

Syllabus

Capital Structure: Debt-equity, Free cash flow, Optimum Capital Structure, Nature of Assets and Capital Structure.

Learning Objective

Format -1	Capital Strutcure
Format -2	NI – Approach
Format -3	NOI – Approach/M-M Approach
Format -4	Trading - Approach
Format -5	Arbitrage Process

Question Analysis – Math Section

2016	2017	2018	2019	2020	2021	2022	2023	2024
Part B: 7 Part C: 13	Part B: Nill Part C: 16	Part B: Nill Part C: 13	Part B: Nill Part C: 13	Part B: 7 Part C: 14	Part B: Nill Part C: 14	Part B: Nill Part C: 13	Part B: Nill Part C: Nill	Part B: 8 Part C: 15
Total = 14	Total = 10	Total = 10	Total = 10	Total = 14	Total = 10	Total = 10	Total = 00	Total = 14



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Format -1: Capital Strutcure

এই ফরমেটের বৈশিষ্ট্যসমূহঃ-

- প্রশ্নপত্রে সরাসরি Earning Per Share (EPS) এর প্রয়োজনীয় তথ্য দেওয়া থাকবে।

প্রিয় ছাত্র-ছাত্রীরা বন্ধুরা, এই অধ্যায়ের অংক সমাধান করার ধাপগুলো নিচে দেখানো হলোঃ-

Working -1: Calculation of Number of Share (NS): -

Alternative	Old	New	Total
A -i			
A-ii			
A-iii			

Working -2: Calculation of interest: -

Alternative	Old	New	Total
A -i			
A-ii			
A-iii			

Working -3: Calculation of Preference Dividend (PD): -

Alternative	Old	New	Total
A -i			
A-ii			
A-iii			

Required -A: Calculation of Earning per shares (EPS): -

Particulars	A -i	A-ii	A-iii
EBIT	000	000	000
(-) interest	000	000	000
EBT	000	000	000
(-) Tax @ --%	000	000	000
EAT	000	000	000
(-) PD	000	000	000
EACS (A)	000	000	000
NS (B)	000	000	000
EPS (A/B)	000	000	000



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Required -B: Calculation of Cost of Equity (Ke): -

$$K_e = \frac{EACS}{\text{Equity Capital}} \times 100$$

Required -C: Calculation of Return of Investment (ROI): -

$$ROI = \frac{EBIT}{\text{Total Investment (old+New)}} \times 100$$

Required -D: Calculation of Indifference Point of EBIT(IP): -

Let, Indifference EBIT = X

We know that, IP: $\left(\frac{(x - i_1)(1 - T) - PD}{NS_1} \right) = \left(\frac{(x - i_2)(1 - T) - PD}{NS_2} \right)$

So, Indifference EBIT = TK ----

Varification Table

Particulars	A -i	A-ii
EBIT	000	000
(-) interest	000	000
EBT	000	000
(-) Tax @ --%	000	000
EAT	000	000
(-) PD	000	000
EACS (A)	000	000
NS (B)	000	000
EPS (A/B)	000	000



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Problem -1: [Commerce Publication]

Following is the capital structure of Sonali Ltd.

Soruce	TK
5,000 Share @ TK 1,000	50,00,000
12% Debt	30,00,000
Total	80,00,000

To Finance an expansion program, the company needs additional capital of TK 20,00,000. There are 3

Alternative:

- (i) Through 12% debt Financing.
- (ii) Through 10% preferred Stock Financing.
- (iii) Selling common share.

Assuming 40% corporate Tax and Expected EBIT of TK 10,00,000.

- (a) Which alternative of financing company should be selected and why?
- (b) Find out the level of EBIT at which the Financial Manager would be indifferent between Altrnative (i) and (iii)

Solution

Working -1: Calculation of Number of Share (NS): -

Alternative	Old	New	Total
A -i	5,000	Nil	5,000
A-ii	5,000	Nil	5,000
A-iii	5,000	$(20,00,000/1,000) = 2,000$	7,000

Working -2: Calculation of Interest: -

Alternative	Old	New	Total
A -i	$(30,00,000 \times 12\%) = 3,60,000$	$(20,00,000 \times 12\%) = 2,40,000$	6,00,000
A-ii	3,60,000	Nil	3,60,000
A-iii	3,60,000	Nil	3,60,000

Working -3: Calculation of Preference Dividend (PD): -

Alternative	Old	New	Total
A -i	Nil	Nil	Nil
A-ii	Nil	$(20,00,000 \times 10\%) = 2,00,000$	2,00,000
A-iii	Nil	Nil	Nil



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Required -A: Calculation of Earning per shares (EPS): -

Particulars	A -i	A-ii	A-iii
EBIT	10,00,000	10,00,000	10,00,000
(-) interest	<u>6,00,000</u>	<u>3,60,000</u>	<u>3,60,000</u>
EBT	4,00,000	6,40,000	6,40,000
(-) Tax @ 40%	<u>1,60,000</u>	<u>2,56,000</u>	<u>2,56,000</u>
EAT	2,40,000	3,84,000	3,84,000
(-) PD	<u>000</u>	<u>2,00,000</u>	<u>000</u>
EACS (A)	2,40,000	1,84,000	3,84,000
NS (B)	<u>5,000</u>	<u>5,000</u>	<u>7,000</u>
EPS (A/B)	<u>48</u>	<u>36.8</u>	<u>54.86</u>

Comment: The company should selected alternative (iii) because int's EPS is higher than other.

Required -B: Calculation of Indifference Point of EBIT(IP): -

$$\text{We know that, } IP = \left(\frac{(EBIT - i_1)(1-T) - PD}{NS_1} \right) = \left(\frac{(EBIT - i_3)(1-T) - PD}{NS_3} \right)$$

$$\text{or, } \left(\frac{(EBIT - 6,00,000)(1-.40) - 0}{5,000} \right) = \left(\frac{(EBIT - 3,60,000)(1-.40) - 0}{7,000} \right)$$

$$\text{or, } \frac{.60EBIT - 3,60,000}{5,000} = \frac{.60EBIT - 2,15,000}{7,000}$$

$$\text{or, } \frac{.60EBIT - 3,60,000}{5} = \frac{.60EBIT - 2,15,000}{7}$$

$$\text{or, } 4.2EBIT - 25,20,000 = 3EBIT - 10,80,000$$

$$\text{or, } 4.2EBIT - 3EBIT = -10,80,000 + 25,20,000$$

$$\text{or, } 1.2EBIT = 14,44,000$$

$$\therefore EBIT = 12,00,000$$

So, Indifference EBIT = TK 12,00,000

Varification Table

Particulars	A -i	A-iii
EBIT	12,00,000	12,00,000
(-) interest	<u>6,00,000</u>	<u>3,60,000</u>
EBT	6,00,000	8,40,000
(-) Tax @ 40%	<u>2,40,000</u>	<u>3,36,000</u>
EAT	3,60,000	5,04,000
(-) PD	<u>000</u>	<u>000</u>
EACS (A)	3,60,000	5,04,000
NS (B)	<u>5,000</u>	<u>7,000</u>
EPS (A/B)	<u>72</u>	<u>72</u>



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Now Try to Yourself

Ex -1: [N.U – 2019/2015 (4th Year, Accounting) AT Part: C13/C15]

The capital structure of Asharaf Ltd. are as follows:

Soruce	TK
Common stock	40,00,000
10% Debenture	20,00,000
Total	60,00,000

The company needs additional capital of TK 20,00,000. It can finance the following ways:

- A. Fully by common stock of TK 100 each.
- B. Fully by 12% Preferred Stock.
- C. Fully by 10% Debenture.

If expected EBIT is TK 10,00,000 and tax rate 50%.

Required:

- (a) EPS?
- (b) Indifference points of EBIT of alternative A and C.

[Ans: (a) TK 6.67, TK 4 & TK 7.5 (b) TK 5]

Ex -2: [N.U – 2017 (4th Year, Accounting) AT Part: C16]

Following is the capital structure of A company.

Soruce	TK
12% Debt	30,00,000
Common stock (80,000 shares)	1,28,00,000
Total	1,58,00,000

To Finance an expansion program, the company needs additional capital of TK 40,00,000. There are 3 Alternative:

- (i) Through 14% debt Financing.
- (ii) Selling common shares of TK 160 per share.

Assuming 40% corporate Tax and Expected EBIT of TK 15,00,000.

- (a) Earning per share under each alternative method of financing.
- (b) Indifference Point of EBIT under method (i) and (ii).

[Ans: (a) TK 4.35 & TK 6.51 (b) TK 13.44]

Ex -3: [N.U – 2016 (4th Year, Accounting) AT Part: B7]

Calculate break even point for two alternatives.

1st alternative: Equity capital TK 4,00,000 @ TK 200 each and Debt is zero.

2nd alternaive: Equity capital TK 2,00,000 @ TK 200 each and Debt is TK 2,00,000 at a coupon rate 10%. Tax rate 50%.

[Ans: EBIT TK 40,000]

Help Line

- IP সূত্র ব্যবহার EBIT করে নির্ণয় করতে হবে।



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Ex -4: [N.U – 2014 (4th Year, Accounting) AT Part: B]

Kushal Ltd. has 40,000 common stock of TK 100 each aggregating TK 40,00,000. It requires additional capital of TK 10,00,000. In order to raise such capital, it has three following alternatives.

- (a) Issuing 10,000 common stock TK 100 each.
- (b) 10% Debt of TK 10,00,000.
- (c) 10% Preferred stock @ TK 100 each TK 10,00,000. Corporate tax rate is 40%.

Find out the EPS under each of three alternatives if EBIT TK 10,00,000.

[Ans: EPS TK 12, TK 13.5 & TK 12.5]

Ex -5: [N.U – 2019/2013 (4th Year, Management) IM Part: C11/C14]

Following is the capital structure of ABC Ltd.

Soruce	TK
12% Debenture	5,00,000
Common Stock (TK 100 Each)	10,00,000
Total	15,00,000

The company needs additional capital of TK 5,00,000. It can finance the following ways:

- (i) 100% equity.
- (ii) 50% equity and 50% Debt @ 12%.
- (iii) 100% Debt @ 12%.

If expected EBIT is TK 20,00,000 and tax rate 40%.

- (i) Calculate the EPS and Which one would you prefer.
- (ii) Calculate the indifference point between (i) and (ii).

[Ans: (1) EPS TK 77.60, TK 91.68 & TK 112.80 & Comment: Alternative (iii) would you prefer because it's EPS higher than other (2) TK 7.20]

Ex -6: [N.U – 2021 (4th Year, Accounting) AT Part: C14]

Following is the capital structure of IDLC.

Soruce	TK
10% Debenture	10,00,000
12% Preferred Stock	10,00,000
Common stock (TK 100 per)	30,00,000
Total	50,00,000

The company needs additional capital of TK 20,00,000. It can finance the following ways:

- (i) Fully by common stock of TK 100 each.
- (ii) Fully by 10% Debenture.
- (iii) Fully by 12% Preferred Stock.

If expected EBIT is TK 10,00,000 and tax rate 40%.

Required:

- (a) Which alternative method of Financing should be selected and why?
- (b) Calculate the indifference EBIT and EPS between alternative (i) and (ii).

[Ans: (1) EPS TK 8.40, TK 10 & TK 5.33 & Comment: Alternative (ii) should be selected because it's EPS higher than other (2) TK 6]



Format -2: NI - Approach

এই ফরমেটের বৈশিষ্ট্যসমূহ:-

- প্রশ্নপত্রে সরাসরি NI Approach এর প্রয়োজনীয় তথ্য দেওয়া থাকবে।

$S = \left(\frac{EBIT - I}{k_e} \right) \text{ or } \left(\frac{EBT}{k_e} \right)$ $B = \left(\frac{I}{k_d} \right)$ $V = S + B$ $P = \left(\frac{S}{NS} \right)$ $k_0 = \left(\frac{EBIT}{V} \times 100 \right)$ Or, $k_0 = k_e + (k_e - k_d) \times \frac{B}{S}$	Here, S = Market value of Share Capital $EBIT$ = Earning before interest and Tax I = Interest K_e = Cost of Equity k_d = Cost of Debt B = Bond K_o = Overall cost of capital V = value of the firm NI Approach = Net Income approach. P = Market Price Per Share																
Calculation of Optimum capital Structure (K_0) <table> <tr> <td>EBIT</td> <td>000</td> </tr> <tr> <td>Less: Interest</td> <td>000</td> </tr> <tr> <td>EBT (NI)</td> <td>000</td> </tr> <tr> <td>K_e</td> <td>000</td> </tr> <tr> <td>$S = EBT/K_e$</td> <td>000</td> </tr> <tr> <td>B</td> <td>000</td> </tr> <tr> <td>$V = S+B$</td> <td>000</td> </tr> <tr> <td>$K_o = (EBIT/V) * 100$</td> <td>000</td> </tr> </table>	EBIT	000	Less: Interest	000	EBT (NI)	000	K_e	000	$S = EBT/K_e$	000	B	000	$V = S+B$	000	$K_o = (EBIT/V) * 100$	000	
EBIT	000																
Less: Interest	000																
EBT (NI)	000																
K_e	000																
$S = EBT/K_e$	000																
B	000																
$V = S+B$	000																
$K_o = (EBIT/V) * 100$	000																



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Now Try to Yourself

Ex -1: [N.U – 2020/2016 (4th Year, Accounting) AT Part: C14/C13]

Assuming no taxes and given the earning before interest and taxes interest at 10% and equity capitalization rate below:

Firm	EBIT	Interest (TK)	Cost of equity
A	2,50,000	25,000	12%
B	3,00,000	60,000	16%
C	5,50,000	2,20,000	15%

Calculate:

- (a) Total market value of each firm.
- (b) Weighted average cost of capital for each firm.

[Ans: V= TK 21,25,000, 21,00,000 & TK 44,00,000 and Ko: 11.76%, 14.29% & 12.50%]

Ex -2: [N.U – 2020 (4th Year, Accounting) AT Part: B7]

You are supplied the following information of Progotio Limited.

Cost of equity 15%, cost of debt 12%, Fixed financial cost TK 24,000 and Earning for the year TK 2,50,000.

Required: (1) Market value of Share (2) Market value of Debenture (3) Total value of the firm (4) overall cost of capital.

[Ans: (1) TK 15,06,667 (2) TK 2,00,000 (3) TK 17,06,667 (4) 14.65%]

Ex -3: [N.U – 2014 (4th Year, Accounting) AT Part: C]

You are supplied the following information of Akib Limited.

Cost of equity 15%, cost of debt 10%, Fixed financial cost TK 20,000 and Earning for the year TK 2,20,000.

Required: (1) Market value of Share (2) Market value of Debenture (3) Total value of the firm (4) overall cost of capital.

[Ans: (1) TK 13,33,333 (2) TK 2,00,000 (3) TK 15,33,333 (4) 14.35%]

Ex -4: [N.U – 2013 (4th Year, Accounting) AT Part: C]

Assuming no taxes and given the earnings before interest and taxes (EBIT), interest at 10% and equity capitalization rate below, calculate the total market value of each firm:

Firms	EBIT	I	Ke (%)
X	2,00,000	20,000	12
Y	3,00,000	60,000	16
Z	5,00,000	2,00,000	15
W	6,00,000	2,40,000	18

Also, determine the weighted average cost of capital for each firm.

[Ans: V= TK 17,00,000, TK 21,00,000, TK 40,00,000 & TK 44,00,000 and Ko: 11.76%, 14.29%, 12.5% & 13.64%]



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Ex -5: [N.U – 2019 (4th Year, Accounting) IM Part: B]

From the following selected data, determine the value of the firms P and Q belonging to the homogeneous risk class, under the NI approach:

Particulars	Leverage firm P	Unleverage firm Q
EBIT	2,25,000	2,25,000
Interest @ 15%	75,000	Nil

Cost of equity capital 20%.

Which of the two firms has an optimum capital structure under the NI approaches?

[Ans: Ko: 18% & 20%% Comment: Leverage firm P]

Help Line

1. $I = \text{TK } 75,000$
2. $B = (75,000/0.15) = \text{TK } 5,00,000$

Ex -6: [N.U – 2015 (4th Year, Accounting) IM Part: B]

Following data of G Ltd. are available: -

$K_e = 16\%$, $K_d = 12.50\%$, Debt = TK 5,75,000, $V_u = \text{TK } 17,18,750$ and Tax rate = 45%.

You are required to calculate: (1) EAT (2) EBIT.

[Ans: (1) TK 2,75,000 (2) TK 5,00,000]

Help Line

1. $EAT = V_u * K_e$
2. $EBIT = (EAT/1-T)$

Ex -7: [N.U – 2017 (4th Year, Management) IM Part: B]

Taj Co. Ltd. has TK 15,00,000 debt capital in its capital structure. Their expected Net operating income is TK 5,00,000. Rate of interest is 6%, overall cost of capital is 12%. The company has 5,000 equity shares.

Required:

- i. Market value of the firm.
- ii. Market value of the equity.
- iii. Market price per equity.

[Ans: (1) TK 41,66,667 (2) 26,66,667 (3) TK 533]

Ex -8: [Commerce Publication]

XYZ Ltd. net operating income TK 2,80,000 weighted average cost of capital is 10%. XYZ Ltd. has got a 5% debt of TK 1,00,000. Find out the value of equity share.

[Ans: TK 27,00,000]

Help Line

1. $V = (EBIT/K_o)$ নির্ণয় করে তারপর $S = V - B$ নির্ণয় করতে হবে।

Ex -9: [N.U – 2020 (4th Year, Management) IM Part: B]

The EBIT of Padma Co. is TK 15,00,000. The interest amount of Debt capital is TK 60,000, cost of equity 16% and cost of Debt 12.5%.

You are required to calculate- (1) Value of the Equity (2) Value of the Company (3) Overall cost of Capital.

[Ans: (1) TK 90,00,000 (2) TK 94,80,000 (3) TK 15.82%]



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Ex -10: [Commerce Publication]

Yarun company with net operating earnings (EBIT) of TK 30,000 is attempting to evaluate a number of possible capital structure, given below which of the capital structure will your recommend and why?

Capital Sturcture	Debt capital	Kd (%)	Ke (%)
1	30,000	10	12
2	40,000	10	12.50
3	50,000	11	13.50
4	60,000	12	15
5	70,000	14	16

[Ans: (1) 11.76%, 12.10%, 12.96%, 14.15% and 15.29% (2) capital Structure 1]

Help Line

2. NI Approach এ ছকের মাধ্যমে K_0 অংকের সমাধান করতে হবে।

Ex -11: [N.U – 2016 (4th Year, Management) IM Part: C]

A company with net operating earnings (EBIT) of TK 3,00,000 is attempting to evaluate a number of possible capital structure, given below which of the capital structures will you recommended and why?

Capital Sturcture	Debt in capital Structure	Kd (%)	Ke (%)
1	3,00,000	10	12
2	4,00,000	10	12.50
3	5,00,000	11	13.50
4	6,00,000	12	15
5	7,00,000	14	16

[Ans: (1) 11.76%, 12.10%, 12.96%, 14.15% and 15.29% (2) capital Structure 1]



Format -3: NOI – Approach/MM Approach

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- প্রশ্নপত্রে সরাসরি MM Approach এর প্রয়োজনীয় তথ্য দেওয়া থাকবে।

For Unleverage Firm [only Share Capital (S) but no Debt (D=0)]	For Leverage Firm [Share capital (S)+ Debt Capital (D)]
$V_u / S = \frac{EBIT(1-T_c)(1-T_s)}{K_e}$ $K_0 = K_e$ মনে রাখতে হবে, Unleverage এ Debt capital বলতে কিছুই নাই। তাই $S=V$ এবং $k_0=k_e$। D, I, Kd, Td এগুলো মান সবময়শ গুণ্য হবে।	(1) $V_L = V_u + \left[1 - \frac{(1-T_c)(1-T_s)}{(1-T_d)} \right] \times D$ Or, $V_L = V_u + (D \times T_c)$ Or, $V_L = V_u + \text{Gain from Leverage} - \text{cost of financial Distress}$ (2) $\text{Gain From Leverage} = \left[1 - \frac{(1-T_c)(1-T_s)}{(1-T_d)} \right] \times D$ Or, $\text{Gain from Leverage} = (D \times T_c)$ Or, $\text{Gain from leverage} = V_L - V_u$ (3) $\text{Cost of Financial Distress} = \text{Debt} \times \text{Rate}$ (4) $K_e = K_0 + (K_0 - K_d)(1-T_c) \times \frac{B}{S}$ [ke=ko] (5) $S = V - B$ (6) $K_0 = \frac{B}{S} \times K_d(1-T_c) + \frac{S}{V} \times K_e$

Here,

S = Market value of Share Capital

EBIT= Earning before interest and Tax

K_e = Cost of Equity

k_d = Cost of Debt

D/B= Debt

V_u = value Unleverage firm

V_L = value leverage firm

T_c = corporate tax rate/Tax rate

T_s = Personal tax rate

T_d = Interest tax rate



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Basic Question -1: [N.U – 2013 (3rd Year, Accounting) FM Part: C]

Following information are available for firm L and U.

Particulars	L	U
EBIT	2,20,000	2,20,000
10% Debt	7,00,000	Nil
K_{eu}		22%
Corporate tax	35%	35%
Personal tax	22%	22%
Interest tax	22%	22%
Cost of financial distress	4.50%	

Requirement: (1) Value of U (2) Gain from Leverage (3) cost of financial distress (4) Value of L

Solution

Req.-1: Calculation of value of unleveraged firm:

$$V_u = \frac{EBIT(1-T_c)(1-T_s)}{K_e} = \frac{2,20,000(1-0.35)(1-0.22)}{0.22} = \text{TK } 5,07,000$$

Req.-2: Calculation of Gain from Leverage:

$$\text{Gain From Leverage} = \left[1 - \frac{(1-T_c)(1-T_s)}{(1-T_d)} \right] \times D = \left[1 - \frac{(1-0.35)(1-0.22)}{(1-0.22)} \right] \times 7,00,000 = \text{TK } 2,45,000$$

Req.-3: Calculation of cost of financial distress:

$$\text{Cost of Financial Distress} = \text{Debt} \times \text{Rate} = 7,00,000 \times 4.50\% = \text{TK } 31,500$$

Req.-4: Calculation of value of leverage firm:

$$V_L = V_u + \text{Gain from Leverage} - \text{cost of financial Distress} = 5,07,000 + 2,45,000 - 31,500 = \text{TK } 7,20,500$$

Basic Question -2: [N.U – 2016 (3rd Year, Accounting), 2016 (4th Year, Management) FM Part: C]

Following information are available for firm L and U.

Particulars	U	L
EBIT	2,50,000	2,50,000
10% Debt	Nil	7,50,000
K_{eu}	15%	
Corporate tax	40%	40%
Personal tax	10%	10%

Requirement:

1. Value of unleveraged firm.
2. Value of leverage firm.
3. Gain from leverage.
4. Cost of equity of unlevered firm.
5. Cost of equity of levered firm.



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Solution

Req.-1: Calculation of value of unleveraged firm:

$$V_u = \frac{EBIT(1-T_c)(1-T_s)}{K_e} = \frac{2,50,000(1-0.40)(1-0.10)}{0.15} = TK \ 9,00,000$$

Req.-2: Calculation of value of leverage firm:

$$V_L = V_u + \left[1 - \frac{(1-T_c)(1-T_s)}{(1-T_d)} \right] \times D = 9,00,000 \left[1 - \frac{(1-0.40)(1-0.10)}{(1-0)} \right] \times 7,50,000 = TK12,45,000$$

Req.-3: Calculation of Gain from Leverage:

$$\text{Gain from leverage} = V_L - V_u = 12,45,000 - 9,00,000 = TK \ 3,45,000$$

Req.-4: Calculation of cost of equity unlevered firm:

$$K_e = 15\% \text{ [since } K_e = K_0 \text{ for unlevered firm]}$$

Req.-4: Calculation of cost of equity levered firm:

$$\begin{aligned} K_e &= K_0 + (K_0 - K_d)(1-T_c) \times \frac{B}{S} \\ &= 0.15 + (0.15 - 0.10)(1-0.40) \times \frac{7,50,000}{4,95,000} \\ &= TK \ 19.55\% \text{ (Ans.)} \end{aligned}$$

$$\begin{aligned} B &= 7,50,000 \\ S &= V - B \\ &= 12,45,000 - 7,50,000 \\ &= TK \ 4,95,000 \end{aligned}$$

Now Try to Yourself

Ex -1: [N.U – 2018 (4th Year, Accounting) AT Part: C13]

A Ltd. and B Ltd. are two different concerns and both of them have a annual operation income of TK 1,20,000. Cost of equity of A is 10% has got no debt while B has got a 5% debt of TK 6,00,000. Corporate tax rate 50%.

Following M-M approach you are required to find out:

1. Total market value of both the firm.
2. Total market value of share both the firm.
3. Cost of equity of both the firm.
4. WACC of both the firm.

[Ans: (1) TK 6,00,000 & TK 9,00,000 (2) TK 6,00,000 & TK 3,00,000 (3) 10% & 15% (4) 10% & 6.67%]

Help Line

- $K_e = K_0 + (K_0 - K_d)(1-T_c) \times \frac{B}{S}$
- $K_{ol} = \frac{B}{V} \times K_d (1-T) + \frac{S}{V} \times K_e$



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Ex -2: [N.U – 2014 (3rd Year, Accounting) FM Part: B]

Green company presently is an unlevered firm. The company expects to generate TK 2,00,000 in earning before interest and taxes (EBIT) in perpetuity. The corporate tax rate is 35%. All earning after taxes is paid as dividends. The firm is considering a capital structure to allow TK 1,00,000 of debt. If cost of debt is 12%. Unlevered firms in the same industry have a cost of equity capital of 18%. What will be the new value of green company?

[Ans: TK 7,57,222]

Help Line

1. $V_u = \frac{EBIT (1-T_c)(1-T_s)}{K_e}$
2. $V_L = V_u + (B \times T_c)$

Ex -3: [N.U – 2019/2014 (4th Year, Accounting) IM Part: C]

Company U and Company L are identical in every respect that U is unlevered, while L has TK 10,00,000 of 5% bonds outstanding.

Assume:

1. That all of the M-M assumptions are met.
2. That there are 40% corporate taxes.
3. That EBIT is TK 2,00,000
4. That the cost of equity to company U is 10%.

Required:

1. Value of U firm.
2. Value of L firm
3. Cost of equity of U firm
4. Market value of share of L firm.
5. Cost of equity of L firm.
6. WACC of U firm.
7. WACC of L firm.

[Ans: (1) TK 12,00,000 (2) TK 16,00,000 (3) 10% (4) TK 6,00,000 (5) 15% (6) 10% (7) 7.5%]



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Ex -4: [N.U – 2018 (3rd Year, Accounting) FM Part: C]

The two companies L = Levered and U = Unlevered, belong to same risk class. The following information are available from the L and U companies:

Particulars	L	U
EBIT	3,30,000	3,30,000
10% Debt	5,00,000	Nil
K_{eu}		22%
Corporate tax	30%	30%
Personal tax	20%	20%
Interest tax	20%	20%
Cost of financial distress	4.50%	

Requirement: (1) Value of U (2) Gain from Leverage (3) cost of financial distress (4) Value of L

[Ans: (1) TK 8,40,000 (2) TK 1,50,000 (3) TK 22,500 (4) TK 9,67,500]

Ex -5: [N.U – 2013 (3rd Year, Accounting) FM Part: B]

There are two companies U and L. Where L is levered firm and U is unlevered firm which has no debt. The firm has TK 10,00,000 Debt at 7%. Assume that there is no corporate tax and personal tax. The expected EBIT is TK 3,00,000 and cost of equity is 16%. There outstanding share 10,000.

Required: (1) Value of U Firm (2) Value of L Firm (3) cost of equity of U firm (4) Market value of share of L Firm.

[Ans: (1) TK 18,75,000 (2) TK 18,75,000 (3) 16% (4) TK 8,75,000]

Ex -6: [N.U – 2013 (4th Year, Accounting) AT Part: B]

You are supplied with the following information. Calculate the value of the firm under MM model.

Earnings before interest and tax TK 6,00,000.

Return on investment (K_e) 15% and tax rate 40%.

[Ans: TK 24,00,000]



Format -4: Traditional Approach

এই ফরমেটের বৈশিষ্ট্যসমূহঃ-

- প্রশ্নপত্রে সরাসরি Traditional Approach এর প্রয়োজনীয় তথ্য দেওয়া থাকবে।

এই ফরমেটের অংকে ২টি প্রশ্ন করা হতে পারে তা নিম্নরূপঃ-

- Find out the overall cost of capital (K_0).
- Find out the optimum capital Structure.

যখন প্রশ্নে Debt and Equity capital এর টাকার পরিমাণ দেওয়া থাকবেঃ-

$$K_0 = \frac{(S \times K_e) + (D \times K_d)}{V} \times 100$$

Comment: Option 2 has optimum capital structure, because it's overall cost of capital is the lower than all other capital structures.

যখন প্রশ্নে Debt and Equity capital এর অনুপাত (%) দেওয়া থাকবেঃ-

$$K_0 = K_d W_d + K_e W_e$$

Comment: Option 2 has optimum capital structure, because it's overall cost of capital is the lower than all other capital structures.

Now Try to Yourself

Ex -1: [N.U – 2015 (4th Year, Accounting) IM Part: B]

In considering the most desirable capital structure of a company, the following estimates of the cost of debt and equity capital (of the tax) have been made at various level of debt:

Debt as a percentage of total capital employed	Cost of Debt (%)	Cost of Equity (%)
0	5	12
10	5	12
20	5	12.5
30	5.5	13
40	6	14
50	6.5	16
60	7	20

Determine the optimum capital structure.

[Ans: (1) 12%, 11.30%, 11%, 10.75%, 10.80%, 11.25% and 12.20% (2) capital Structure 4]



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Ex -2: [N.U – 2013 (4th Year, Accounting) IM Part: C]

You are provided the probable capital structures of a company.

Situation	Debt	Equity	After tax cost of debt	Cost of equity
1	5,00,000	3,00,000	8%	10%
2	2,50,000	5,50,000	7%	11%
3	1,00,000	7,00,000	6%	12%

Find out the optimum capital structure of the company.

[Ans: (1) 8.75%, 9.75%, 11.25% (2) Situation 1]

Ex -3: [Commerce Publication]

The Green Dreem Ltd. going declare the most desirable capital structure of the company various level of debt equity mix (after tax) is given below:

Debt as a percentage of total capital employed	Cost of Debt (%)	Cost of Equity (%)
Mr. A - 0	5	12
Mr. B - 10	5	12
Mr. C - 15	5	12.5
Mr. D - 20	5.5	13
Mr. P - 30	6	14
Mr. Q - 40	6.5	16
Mr. R - 50	7	20

Determine the optimum Debt equity mix for the company by calculating composite cost of capital.

[Ans: (1) 12%, 11.30%, 11.38%, 11.50%, 11.60%, 12.20% and 13.50% (2) Comment:

Optimum Debt equity mix is 10%: 90% because in this level overall cost is lowest]

Ex -4: [N.U – 2016 (4th Year, Accounting), 2014 (3rd Year, Accounting) Part: C]

ABC company with net operating earnings (EBIT) of TK 3,00,000 and Total capital of TK 10,00,000 is attempting to evaluate a number of possible capital structure, given below which of the capital structures will you recommended and why?

Capital Sturcture	Debt in capital Structure	Kd (%)	Ke (%)
1	3,00,000	10	12
2	4,00,000	10	12.50
3	5,00,000	11	13.50
4	6,00,000	12	15
5	7,00,000	14	16

[Ans: (1) 11.40%, 11.50%, 12.25%, 13.20% and 14.60% (2) capital Structure 1]

Ex -5: [N.U – 2014 (4th Year, Accounting), 2014 (4th Year, Management) Part: C]

A firm wants to determine its opital structure. On the basis of following information regarding Debt and Equity and their respective cost. Calculate the optimal capital structure of the firm:-

Situation	Debt	Equity	Cost of Debt	Cost of Equity
1	4,00,000	6,00,000	8%	7%
2	3,00,000	7,00,000	10%	6%
3	6,00,000	4,00,000	12%	9%

[Ans: (1) 7.40%, 7.2% and 10.80% (2) capital Structure 2]



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Ex -6: [N.U – 2019 (4th Year, Management) Part: C]

ABC company the following debt and equity capital (After tax) have been made at various investment plans. You are required to determine optimum debt equity mix for the company by calculating the composite cost of capital:

Options	% of Debt	Cost of Debt (%)	Cost of Equity (%)
A	0	0	12
B	10	5	12
C	20	5.2	12.50
D	30	6	13
E	40	6.5	14
F	50	7	15
G	60	7.5	16

[Ans: (1) 12%, 11.30%, 11.04%, 10.90%, 11%, 11% and 10% (2) capital Structure D]



Format -5: Arbitrage Process

এই ফরম্যাটের বৈশিষ্ট্যসমূহ

- একজন বিনিয়োগকারীর সামনে দুই ধরনের বিনিয়োগের সুযোগ বিদ্যমান।
- একটি হলো লিভারযুক্ত ফার্মে (Levered Firm) বিনিয়োগ এবং অপরটি হলো লিভারজমুক্ত ফার্মে (Unlevered Firm) বিনিয়োগ।
- Levered ফার্মে বিনিয়োগ করলে মোট বিনিয়োগ হবেঃ-
 $\text{Total Investment} = \text{Debt} + \text{Equity}$
 মোট আয় হবেঃ-
 $\text{Total Income} = \text{Interest} + \text{Dividend}$
- ফার্মে বিনিয়োগ করলে মোট বিনিয়োগ হবেঃ-
 $\text{Total Investment} = \text{Only Equity}$
 মোট আয় হবেঃ-
 $\text{Total Income} = \text{Dividend}$
- Interest নির্ণয় করতে হবে Debt capital এর উপর এবং Dividend নির্ণয় করতে হবে Net income এর উপর।
- সর্বশেষ দেখতে হবে যদি দুটি ফার্মেই বিনিয়োগকারীর মোট আয় সমান হয় তাহলে সে ফার্মের তার মোট বিনিয়োগ কম সে ফার্মেই বিনিয়োগকারী চলে যাবে (Switch করবে)

Current Position (L)	TK	Position will be (U)	TK
Total Investment		Total Investment:	
Equity Capital (S*Investors rate)	000	Equity Capital (S*Investors rate)	000
Debt Capital (D*Investors rate)	000		000
	000		
Total Income:		Total Income:	
Dividend (Net income * investors rate)	000	Dividend (Net income * investors rate)	000
Interest (Debt capital * interest rate)	000		000
	000		000

Comment:

Req- a: The investor will reduce his outlay by TK ---- through the use of Leverage.

Req-b: According to MM, this arbitrage process will come to an end when the value of both the firm same.

- যদি একজন বিনিয়োগকারী কোন ফার্মে কত % বিনিয়োগ করেছেন তা প্রশ্নে সরাসরি উল্লেখ না থাকে তাহলে প্রথমেই তার বিনিয়োগের % বের করে নিতে হবে নিম্নরূপ ভাবেঃ-
- যদি বিনিয়োগকারীর বিনিয়োগের পরিমাণ “টাকায়” দেওয়া থাকে তখনঃ-

$$\% \text{ of investors} = \frac{\text{Investment TK}}{\text{Total Equity capital of the firm}} \times 100$$

- যদি বিনিয়োগকারীর বিনিয়োগের পরিমাণ “শেয়ারে” দেওয়া থাকে তখনঃ-

$$\% \text{ of investors} = \frac{\text{Investment No. of Share}}{\text{Total No. of Share of the firm}} \times 100$$



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Now Try to Yourself

Ex -1: [N.U – 2015 (4th Year, Accounting) IM Part: C]

There are two firms L and U which are identical in all respects except that firm L has 10% TK 5,00,000 debentures. The EBIT of both the firm is TK 1,00,000. The equity capitalization rate of the levered firm is slightly higher 15% than that of the unlevered firm 13%.

Requirement:

1. Determine the total value and overall cost of capital of the two firms.
2. Mr. Hossain an investor holds 10% of the outstanding shares in firm X. Should be prefer swithing from X to Y?

[Ans: (1) Value TK 8,33,333 & TK 7,69,231 and 12% & 13% (2) The investor will reduce his outlay by TK 6,410 through the use of Leverage.]

Ex -2: [N.U – 2019 (4th Year, Management) FM Part: C]

There are two X and Y which are identical in all respects except that firm X has 10% TK 5,00,000 debentures. The EBIT of both the firm is TK 5,00,000. The equity capitalization rate of the levered firm is slightly higher 14% than that of the unlevered firm 12%.

Requirement:

1. Determine the total value and overall cost of capital of the two firms.
2. Mr. Hossain an investor holds 10% of the outstanding shares in firm X. Should be prefer swithing from X to Y?

[Ans: (1) Value TK 37,14,286 & TK 41,66,667 and 13.46% & 12% (2) The investor will reduce his outlay by TK 45,238 through the use of Leverage.]

Ex -3: [N.U – 2017 (4th Year, Management) FM Part: C16]

The following is the data regarding two companies SUN and MOON belonging to the same risk class:

Particulars	SUN	MOON
EBIT	3,00,000	3,00,000
6% Debt		10,00,000
Cost of Equity	10%	15%

Requirement:

1. Value of the firms.
2. Overall cost of capital.
3. Assume that an investor owns 10% of SUN company's share. Using Arbitrage Process show that, how can he obtain same return at a lower cost.

[Ans: (1+2) Value TK 30,00,000 & TK 26,00,000 and 10% & 11.54% (3) The investor will reduce his outlay by TK 40,000 through the use of Leverage.]



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Ex -4: [N.U – 2019 (4th Year, Management) IM Part: C12]

The following is the data regarding two companies P and Q belonging to the same risk class:

Particulars	P	Q
EBIT	2,40,000	2,40,000
10% Debt	6,00,000	3,00,000
Cost of Equity	12%	12.5%

Requirement:

1. Value of the firm P.
2. Value of the firm Q.
3. Assume that an investor owns 10% of P's share. How can the investor obtain same return at a lower cost?
4. When will the arbitrage process come to an end?

[Ans: (1) TK 21,00,000 (2) TK 19,80,000 (3) The investor will reduce his outlay by TK 12,000 (4) Comment: According to MM, this arbitrage process will come to an end when the value of both the firm same.]

Help Line

Arbitrage Process

Current Position	TK	Position will be	TK
Total Investment		Total Investment:	
Equity Capital (15,00,000*10%)	1,50,000	Equity Capital (16,80,000*10%)	1,68,000
Debt Capital (6,00,000*10%)	60,000	Debt Capital (3,00,000*10%)	30,000
	2,10,000		1,98,000
Total Income:		Total Income:	
Dividend (1,50,000 * 12%)	18,000	Dividend (1,68,000 * 12.50%)	21,000
Interest (60,000 * 10%)	6,000	Interest (30,000 * 10%)	3,000
	24,000		24000



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Ex -5: [N.U – 2016 (4th Year, Accounting), 2016 (4th Year, Management) Part: C]

The Kelly Co and Green Co. are identical in every respect that except that Kelly Co. is not levered. While the Green Co has TK 30,00,000 @ 10% bonds outstanding. There are no taxes and capital markets are assumed to be perfect. The valuation of two firms in the following: -

Particulars	Kelly	Green
Net operating income (EBIT)	7,50,000	7,50,000
Interest on Debt (I)	000	3,00,000
Earnings to equity holders (NI)	7,50,000	4,50,000
Equity capitalization rate (Ke)	0.15	0.20
Market value of equity (S)	50,00,000	22,50,000
Market value of Debt (B)	000	30,00,000
Total market value of the firm (V= S+B)	50,00,000	52,50,000
Implied overall capitalization rate (Ko)	15%	14.33%
Debt equity rate (B/S)	0	1.33

If you own TK 2,25,000 share capital of company green stock.

1. Show the net income from arbitrage.
2. The amount by Which you could reduce your outlay throughout the use of arbitrage?

[Ans: (1) Net Income TK 75,000 (2) The investor will reduce his outlay by TK 25,000 through the use of Leverage]

$$\% \text{ of investors} = \frac{\text{Investment TK}}{\text{Total Equity capital of the firm}} \times 100 = \frac{2,25,000}{22,50,000} \times 100 = 10\%$$

Ex -6: [N.U – 2014 (4th Year, Management) Part: C]

Two firms A and B are identical in all respect expect that B has TK 5,00,000 debt outstanding at a 6% rate of interest. The values of the two firms are as follows:

Particulars	Firm A	Firm B
Net operating income (EBIT)	1,50,000	1,50,000
Interest on Debt (I)	000	30,000
Earnings to equity holders (NI)	1,50,000	1,20,000
Equity capitalization rate (Ke)	0.10	0.15
Market value of equity (S)	15,00,000	8,00,000
Market value of Debt (B)	000	5,00,000
Total market value of the firm (V= S+B)	15,00,000	13,00,000
Implied overall capitalization rate (Ko)	0.10	0.1154

Assume that an investor holds 10% of the B firm's share. How can the investor obtain same return at a lower cost?

[Ans: (1) The investor will reduce his outlay by TK 20,000 through the use of Leverage.]



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Recent Question

Problem-1: [N.U – 2022 (4th Year, Accounting) AT Part: C13]

The Red Sun Company expects perpetual earnings before interest and taxes of TK 10,00,000 per year. The after-tax rate of return is 10% of all equity firm. Currently the firm has TK 15,00,000 debt subject to corporate tax rate 30%. The personal tax rate on interest income is 15% and the personal tax rate on equity distribution is zero. The combined financial distress and agency cost associated with the debt are approximately 5% of the total value of the debt.

- (a) What is the value of firm?
- (b) What is the added value of firm's debt?

[Ans: (1) $V_u = \text{TK } 70,00,000$ & $V_L = \text{TK } 71,89,706$ (1) added Value of firm's debt = $V_L - V_u = 71,89,706 - 70,00,000 = \text{TK } 1,89,706$]

Problem-2: [N.U – 2024 (4th Year, Accounting) AT Part: B8]

Two alternative capital structure of Neon Company are as follows.

- 1st alternative: Equity capital TK 5,00,000 @ TK 100 each and Debt is zero.
- 2nd alternative: Equity capital TK 3,00,000 and Debt is TK 2,00,000 at a coupon rate 10%.
Tax rate 40%.

Determine the break-even point for two alternatives.

[Ans:]

Help Line

- IP সূত্র ব্যবহার EBIT করে নির্ণয় করতে হবে।

Problem-3: [N.U – 2024 (4th Year, Accounting) AT Part: C15]

Alaskan corporation has earnings before interest and taxes of 3 million and a 40% tax rate. Its required rate of return on equity in the absence of borrowing is 18%.

- (a) In the absence of personal taxes, what is the value of the company in MM Model.
 - (i) With no leverage?
 - (ii) With TK 5 million in debt?
- (b) Personal as well as corporate taxes now exist. The marginal personal tax rate on common stock income is 30% and the marginal personal tax rate on debt income is 25%.

Required:

- (i) Determine the value of the company for each of the two-debt alternative in part (a).
- (ii) Why do your answer differ?

[Ans:]